Learning Management System (LMS) - MySQL Project

Problem Statement
Develop a database for an online learning platform where students enroll in courses, complete modules and take assessments.

Expected Outcome
Display learning progress, completed modules and assessment performance.

Database Design

- Students(student_id PK, student_name, email, phone, city)
- Mentors(mentor_id PK, mentor_name, expertise)
- Courses(course_id PK, mentor_id FK, course_name, category, duration)
- Enrollment(enrollment_id PK, student_id FK, course_id FK, enroll_date, status)
- Assessments(assessment_id PK, student_id FK, course_id FK, score, assessment_date)

---- **STUDENTS (10 Records)**

```
INSERT INTO Students VALUES

(101,'Aarav Sharma', 'aarav@gmail.com','9876500001','Hyderabad'),
(102,'Bhavya Reddy', 'bhavya@gmail.com','9876500002','Vijayawada'),
(103,'Charan Kumar', 'charan@gmail.com','9876500003','Visakhapatnam'),
(104,'Divya Patel', 'divya@gmail.com','9876500004','Bengaluru'),
(105,'Esha Gupta', 'esha@gmail.com','9876500005','Chennai'),
(106,'Farhan Ali', 'farhan@gmail.com','9876500006','Mumbai'),
(107,'Gopi Krishna', 'gopi@gmail.com','9876500007','Warangal'),
(108,'Harini Rao', 'harini@gmail.com','9876500008','Guntur'),
(109,'Ishaan Verma', 'ishaan@gmail.com','9876500009','Pune'),
(110,'John Mathew', 'john@gmail.com','9876500010','Kochi');
```

---- **MENTORS (5 Records)**

```
INSERT INTO Mentors VALUES

(201,'Dr. Santhosh Kumar', 'Machine Learning'),
(202,'Priya Nair', 'Data Analytics'),
(203,'Rahul Mehta', 'Web Development'),
(204,'Sneha Reddy', 'Cloud Computing'),
(205,'Arjun Singh', 'Cyber Security');
```

---- **COURSES (6 Records)**

INSERT INTO Courses VALUES

(301,201,'Machine Learning','AI',60),
(302,202,'Data Analytics','Analytics',45),
(303,203,'Python Full Stack', 'Web Development',75),
(304,204,'AWS Cloud','Cloud',40),
(305,205,'Cyber Security Basics','Security',35),
(306,202,'Power BI', 'Business Intelligence',30);

----- **ENROLLMENTS (12 Records)**

INSERT INTO Enrollment VALUES

(501,101,301,'2026-07-01','Active'),
(502,102,302,'2026-07-02','Completed'),
(503,103,303,'2026-07-03','Active'),
(504,104,304,'2026-07-04','Completed'),
(505,105,305,'2026-07-05','Active'),
(506,106,306,'2026-07-06','Completed'),
(507,107,301,'2026-07-07','Active'),
(508,108,302,'2026-07-08','Completed'),
(509,109,303,'2026-07-09','Dropped'),
(510,110,304,'2026-07-10','Active'),
(511,101,306,'2026-07-11','Completed'),
(512,103,305,'2026-07-12','Active');

----- ASSESSMENTS (15 Records)

INSERT INTO Assessments VALUES

(601,101,301,'Quiz 1',85.50,'2026-07-05'),
(602,101,301,'Final Assessment',91.00,'2026-07-15'),
(603,102,302,'Assignment 1',78.00,'2026-07-06'),
(604,102,302,'Final Assessment',88.50,'2026-07-16'),
(605,103,303,'Quiz 1',81.00,'2026-07-07'),
(606,103,303,'Project Evaluation',75.50,'2026-07-17'),
(607,104,304,'Quiz 1',92.00,'2026-07-08'),
(608,104,304,'Final Assessment',89.00,'2026-07-18'),
(609,105,305,'Assignment 1',84.00,'2026-07-09'),
(610,106,306,'Quiz 1',91.50,'2026-07-10'),
(611,107,301,'Assignment 1',80.00,'2026-07-11'),
(612,108,302,'Quiz 2',87.00,'2026-07-12'),
(613,109,303,'Final Assessment',83.50,'2026-07-13'),
(614,110,304,'Project Evaluation',86.00,'2026-07-14'),
(615,101,306,'Dashboard Project',95.00,'2026-07-20');

SQL Practice Questions

1. List all students.

2. Delete an inactive enrollment.

3. Update the enrollment status of a student from 'Active' to 'Completed'.

4. Display all mentor names in uppercase and course names in lowercase.

5. Show first 5 characters of course names.

6. Concatenate student name and city.

7. Display the last four digits of each student's phone number

8. Replace 'Python' with 'Advanced Python' in output.

9. Find courses containing 'Data'.

10. Display today's assessments.

11. Find the number of days since each student enrolled.

12. Display the assessment date in the format **DD-Mon-YYYY**.

13. Show all enrollments made in July 2026.

14. Count the number of students enrolled in each course.

15. Display the average assessment score for each course.

16. Find the highest and lowest assessment score.

17. Display courses having more than 5 students.

18. Display student names along with their enrolled course names.

19. Display course names with their mentor names.

20. Display student names, course names, and enrollment status.

21. Display assessment details along with student and course names.

22. Display mentor names along with the courses they teach.

23. Students without enrollments.

24. Courses without modules.

25. Modules not yet completed.

26. Find students who scored higher than the average assessment score.

27. Display the course having the maximum duration.

28. Correlated subquery for top scorer per course.

29. Create a view to display student names, course names, and enrollment status.

30. Create a view to display assessment results with student and course details.

31. Retrieve all records from both created views.

32. Start a transaction and update the enrollment status of a student.

33. Create a SAVEPOINT after updating assessment scores.

34. Roll back to the SAVEPOINT.

35. Commit the transaction.

36. Delete an assessment record and roll back the transaction.

37. Create a stored procedure to display all courses belonging to a particular category.
Input: category

38. Create a stored procedure to find the average assessment score of a particular student.
Input: student_id

39. Create a trigger on Assessments to store newly added assessment details in an Assessment_Log table.

40. Create an AFTER UPDATE trigger on Enrollment to store the old and new enrollment status in an Enrollment_Status_History table.